LAB 1 (Microprocessor and Assembly Language)

Objective:

1. Overview of Dyna- 85 microprocessor kit and programming model.
2. WAP for Addition of two 8-bit hex numbers.
3. WAP for Subtraction of two 8-bit hex numbers.

THEORY:

SYSTEM HARDWARE OVERVIEW: CENTRAL PROCESSING UNIT: DYNA-85 is based on the INTEL 8085A high performance CPU operating at 3 MHZ.

MEMORY: Powerful system monitor has been provided on a 27c128 EPROM covering 16k bytes. A6264 battery backup RAM (2K) is provided on the board for inputting & executing programs. Three 28 pin sockets are provided for memory chips so that further expansion of RAM/ EPROM is possible up to a maximum of 56K.

HEX KEYPAD/ DISPLAY INTERFACE: A keypad with 21 keys & 6 digits LED 7 segment display is provided for interaction with the system using 8279 keyboard / display controller.

PARALLEL I/O INTERFACE: 46 parallel I/O lines are provided on board, 22 from 8155 & 24 from 8255. these lines are bought on to connector J2 (for 8155) & J3 (for8255).

SERIAL I/O INTERFACE: Serial I/O is available through RS232C compatible port. The SID & SOD lines are used under software control for serial operation. Baud rates auto adjustable.

TIMER: Three channels of 8253 chip, a 16-bit timer/ counter & one channel of 14-bit timer/counter of 8155 are provided on board. All lines of 8253 are provided on connector J7 & lines of 8155 are provided on J2.

SYSTEM COMMANDS OVERVIEW:

The HEX KEYPAD mode supports the following commands:

RESET: Provides hardware reset. Display shows “friend” on pressing this key.

VI: Vector interrupt key. Activates RST 7.5 vectored interrupt.

SET: Allows the user to examine & modify the contents of RAM & only examination of contents is possible in case of EPROM.

INR: Increments memory address presently displayed in the address field of display.

DCR: Decrements memory address presently displayed in the address field of display.

REG: Allows the user to examine contents of CPU registers & modify them it necessary.

GO: Allows the user to load the program counter by the desired memory address which is the starting address of the program to be executed.

EXEC: Used to start the execution of GO or CODE command.

CODE: Used for selecting one of the coded sub routines in the monitor.

STEP: Allows the user to execute the program in single step mode or break point mode.

SAVE: Used for saving the contents of memory onto an audio cassette.

LOAD: Used for loading the program from audio cassette back to memory in RAM area.

U1toU4: These keys are user definable function keys. The functions of these keys can be defined by user, by loading the appropriate memory locations with vectors pointing to user subroutines.

MEMORY MAPPING:

The system memory is also as important as the CPU itself, because this is where the system program resides & the CPU takes its instruction from the program. The memory is of two types ROM & RAM i.e. READ ONLY MEMORY & RANDOM ACCESS MEMORY. The DYNA-85 has a flexible memory map, & for your convenience for program development, the RAM has useful features such as battery back-up.

0000H to 3FFFH: Monitor EPROM

Monitor 2732 is located at 0000H to 0FFFH & is mapped at 1000H-1FFFH,2000H-2FFFH& 3000H-3FFFH also. If 2764/27128 are used, 1000H-3FFFH can be used for further expansion.

4000H to BFFFH: Expansion for EPROM/ RAM

This socket is used for user expansion of EPROM & RAM. EPROMs like2716/2732/2764/27128/17256 or RAMs like 6116/6226/62256 can be installed by suitable strapping.

C000H to FFFFH: User RAM IC 6116.

User RAM sockets. The 2K user RAM IC 6116is located at F800H-FFFFH.this 2K memory is folded after every 2K bytes from C000H to FFFFH. In this socket 6264 chip also can be used.

POWER SUPPLY:

Power supply for DYNA-85 kit should have following specification:

VOLTAGE	CURRENT RATING
+5V	1 A
+12V	500 mA
-12V	250 mA
+30V	100 mA

PROGRAMMING MODEL:

Accumulator A (8) Fl	Flag register
B (8)	C (8)
D (8)	E (8)
H (8)	L (8)
Stack Pointer (SP) (16-bit)	
Program counter (PC) (16-bit)	

D7	D6	D5	D4	D3	D2	D1	D0
S	Z		AC		P		CY

REGISTERS: -

The 8085 has 6 general purpose register to store 8-bit data; these are classified as B, C, D, E, and H & L as shown in fig 1. They can be combined as register pairs –BC, DE, and HL to perform some 16 bit operations. The programmer can use these registers to store or copy data into the registers by using data copying instructions.

ACCUMULATOR: -

The accumulator is an 8-bit resister that is part of arithmetic/ logic unit(ALU). This resister is used to store 8-bit data & to perform arithmetic & logic operations. It can also be used as general purpose register.

FLAGS: -

The ALU includes 5 flip-flops, which are set or reset after an operation according to data conditions of the result in the accumulator & other registers. They are called Zero (Z), Carry (CY), Sign (S), Parity (P), & Auxiliary carry (AC) flags;

Z – Zero: The zero flag is set to 1 when the result is zero; otherwise it is reset

CY – Carry: If an arithmetic operation results in a carry, the CY flag is set; otherwise it is reset

S – Sign: The sign flag is set if bit D7 of the result is 1; otherwise it is reset

P – Parity: If the result has an even number of 1s the flag is set, for odd number of 1s, the flag is reset

AC – Auxiliary carry: In an arithmetic operation, when carry is generated by digit D3and passed to digit D4, AC flag is set

PROGRAM COUNTER: This 16-bit resister deals with sequencing the execution of instructions and works as memory pointer. Memory locations have 16-bit addresses hence this is a16-bit resister. The function of program counter is to point to memory address from which the next byte is to be fetched.

STACK POINTER: The stack pointer is also a 16- bit resister used as a memory pointer. It points to a memory location in R/W memory, called stack. the beginning of a stack is defined by loading a 16-bit address in the stack pointer.

Objective2: WAP for Addition of two 8-bit hex numbers.

Address	Hex Code(opcode)	Mnemonics	Comments
C000	3E H	MVI A, 08H	Loads the accumulator with 08H hex number
C001	08		
C002	06 H	MVI B, 06H	Load the register B with 06H hex number
C003	06		
C004	80 H	ADD B	Add the content of B with A
C005	32 H	STA 2050	Store the content of A on 2050H address.
C006	50		
C007	20		
C008	76 H	HLT(CF)	End

STEPS:

1. Press Reset Button to reset the kit.
2. For entering the address, press the set button then the display is changed to '.' It indicates that the kit is ready to take the address.
3. Enter the starting address, once the starting address is entered**, press INR** key to change from address mode to data mode and enter the data for that particular address.
4. Press INR key again then address is automatically incremented by one and you need to provide data in that particular address.
5. The process is repeated for entire program to provide address and data.
6. To execute the program, press RESET + GO + Starting Address + EXEC Key
7. For getting the output, press RESET+ SET + Given Result Address + INR

Write output:

Objective3: WAP for Subtraction of two 8-bit hex numbers.

Address	Hex Code(opcode)	Mnemonics	Comments
C000	3E H	MVI A, 08H	Loads the accumulator with 08H hex number
C001	08		
C002	06 H	MVI B, 06H	Load the register B with 06H hex number
C003	06		
C004	90 H	SUB B	Add the content of B with A
C005	32 H	STA 2050	Store the content of A on 2050H address.
C006	50		
C007	20		
C008	76 H	HLT(CF)	End